

Chapter 2

Communication and Power Consumption Models

Bibliographic Reference

This chapter is based in part on: E. Peschiera, S. Yun, Y. Lee, L. Van der Perre, and F. Rottenberg, “A parametric power model of upper mid-band (FR3)* base stations for 6G,” in *Proceedings of 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2026)*, Barcelona, Spain, 2026, pp. 21476-21480 [42].¹

*The upper mid-band, also known as Frequency Range 3 (FR3) within the 3rd Generation Partnership Project (3GPP), comprises the carrier frequencies from 7.125 GHz to 24.25 GHz. It is one of the main candidate frequency bands for sixth-generation (6G) networks, mainly to provide a target ten-fold capacity increase in urban areas [43, 44, 45, 46].

In this chapter, we describe the system model used in the thesis, covering the hardware (HW) devices included in the base station (BS) and their power consumption models, and the communication process between the BS and the users. We describe downlink (DL) and uplink (UL) scenarios and adopt a general approach to both communication and power models. In the next

¹The Python simulation code is available in open access at the following link: github.com/dramco-research/fr3-power-model-epeschiera-icassp26.

chapters, we particularize the models presented here to the scenario under study. The proposed power model is a generalization of existing ones in the literature, proposed for BSs working in the sub-6 GHz band, also known as Frequency Range 1 (FR1), or in the millimeter wave (mmWave) band, also known as Frequency Range 2 (FR2).² More details on the differences with respect to the state of the art are given at the beginning of Section 2.2. We discuss its application to prospective upper mid-band (or FR3) BSs in Section 2.3.

Throughout this thesis, we consider a BS equipped with M antennas and power amplifiers (PAs) in DL, and M antennas and low-noise amplifiers (LNAs) in UL. The BS has M_{RF} radio frequency (RF) chains in DL and M_{RF} RF chains in UL.³ The BS serves single-antenna users using space-division multiple access (SDMA), meaning that all users are served on the same time-frequency resources. The transmission and reception adopt orthogonal frequency-division multiplexing (OFDM). The maximum number of supported users is K_{max} , which we consider to be the same in DL and UL, while the number of active users is K_{DL} in DL and K_{UL} in UL. The DL takes place on a carrier frequency $f_{\text{c,DL}}$ and the UL takes place on a carrier frequency $f_{\text{c,UL}}$. Both DL and the UL utilize a bandwidth of B Hz and a subcarrier spacing of Δ_f Hz. The bandwidth is divided into Q_{IFFT} subcarriers, of which Q_{max} are available to transmit/receive data symbols. The OFDM symbol period, including cyclic prefix, is denoted by T . An illustration of the frequency-domain structure considered in this thesis is given in Fig. 2.1. A subcarrier is denoted as ‘active’ when it is allocated non-zero power. Within the bandwidth, there is a number of null subcarriers, and those can be arbitrarily placed in the bandwidth. There are no pilot subcarriers in an active OFDM symbol because we consider, as a simplifying assumption, that the reference signalling is separated in time from the data transmission/reception (cf. Fig. 2.2).

We consider a frame of T_{frame} seconds divided into DL and UL phases. The DL phase has a duration of $\tau_{\text{DL}}T_{\text{frame}}$ seconds, where $\tau_{\text{DL}} \in [0, 1]$, while the UL phase has a duration of $\tau_{\text{UL}}T_{\text{frame}}$ seconds, where $\tau_{\text{UL}} \in [0, 1]$. In DL (UL), there are N_{DL} (N_{UL}) available OFDM symbols,⁴ of which $N_{\text{a,DL}}$ ($N_{\text{a,UL}}$) are activated, i.e., carrying users’ data. By considering $i \in \{\text{DL}, \text{UL}\}$ as the variable representing either the DL or the UL, each phase is further subdivided in three phases:

²FR1 spans the carrier frequencies from 410 MHz to 7.125 GHz, while FR2 spans the carrier frequencies from 24.25 GHz to 71 GHz [47].

³In general, we refer to an “RF chain” as a signal chain from the inverse fast Fourier transform input to the phase shifter input in DL, and viceversa in UL (cf. Fig. 2.4). An RF chain includes, therefore, digital and analog modules. When discussing the digital processing modules specifically, we refer to a digital signal processing (DSP) chain as the part of the RF chain that goes from the inverse fast Fourier transform to the digital-to-analog converter in DL, and from the analog-to-digital converter to the fast Fourier transform in UL.

⁴In the thesis, we interchangeably use the terms OFDM symbol and time slot.

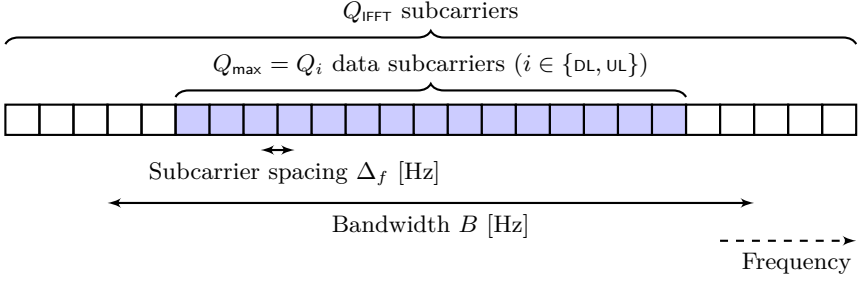


Figure 2.1: Frequency-domain structure considered in this thesis, applicable to both DL and UL. We consider that for an active user and active OFDM symbol, all data subcarriers are utilized, giving $Q_i[k, n] = Q_{\max}$.

- Reference signaling, lasting $\tau_i \tau_{\text{sig}, i}$ seconds, where $\tau_{\text{sig}, i} \in [0, 1]$ is the ratio of DL or UL signaling duration to DL or UL duration. In this phase, the BS and the users exchange synchronization, control, broadcast, and pilot signals used, e.g., to estimate channel state information (CSI) [32].
- Data transmission or reception, lasting $\tau_i(1 - \tau_{\text{sig}, i}) \frac{N_{a,i}}{N_i}$ seconds.
- No reference signaling and no data transmission or reception, lasting $\tau_i(1 - \tau_{\text{sig}, i})(1 - \frac{N_{a,i}}{N_i})$ seconds. This corresponds to micro-sleep mode, where the BS is active but is not transmitting or receiving any signal. In DL, this mode is also known as discontinuous transmission (DTX) [30].

A visual illustration of the time domain subdivision is given in Fig. 2.2.

We define the instantaneous physical resource load in DL or UL of user k and time slot n as the number of utilized data subcarriers over the number of available data subcarriers

$$\ell_i[k, n] = \frac{Q_i[k, n]}{Q_{\max}}, \quad i \in \{\text{DL}, \text{UL}\}. \quad (2.1)$$

for $k = 1, \dots, K_{\max}$ and $n = 1, \dots, N_i$. Averaging over all the users and the time slots, we compute the average physical resource load

$$\bar{\ell}_i = \frac{1}{N_i} \sum_{n=1}^{N_i} \left(\frac{1}{K_{\max}} \sum_{k=1}^{K_{\max}} \frac{Q_i[k, n]}{Q_{\max}} \right), \quad i \in \{\text{DL}, \text{UL}\}. \quad (2.2)$$

To ease the presentation, the set of Active and Non-active users is represented by ordered indexes $\{1, \dots, K_i\}$ and $\{K_i + 1, \dots, K_{\max}\}$, respectively, while the set of active and non-active time slots is represented by ordered indexes $\{1, \dots, N_{a,i}\}$

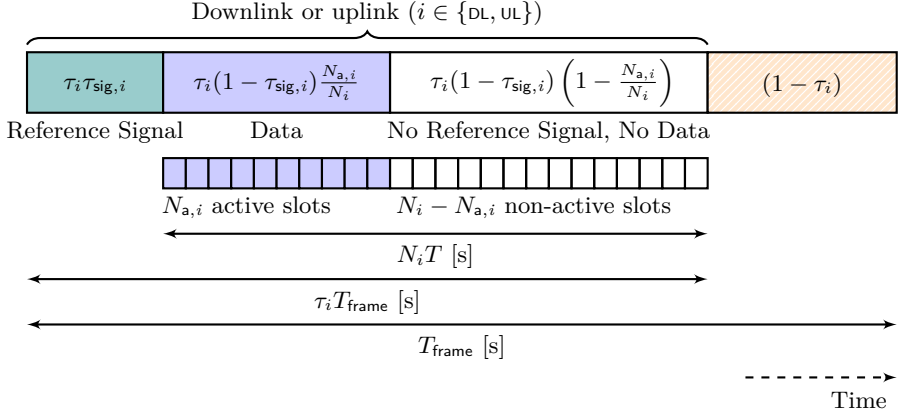


Figure 2.2: Time-domain (frame) structure considered in this thesis, where the quantities inside each rectangle are time ratios relative to the frame duration T_{frame} . In FDD, $\tau_{\text{DL}} = \tau_{\text{UL}} = 1$ and the rightmost rectangle has zero duration.

and $\{N_{a,i} + 1, \dots, N_i\}$, respectively. In this thesis, we assume that when a user is active in an active time slot, i.e., $k \in \{1, \dots, K_i\}$ and $n \in \{1, \dots, N_{a,i}\}$, it is allocated all the available subcarriers giving $Q_i[k, n] = Q_{\text{max}}$. In other words, we do not consider orthogonal frequency-division multiple access (OFDMA). Given this assumption, the average load in equation (2.2) simplifies to

$$\bar{\ell}_i = \frac{N_{a,i}}{N_i} \frac{K_i}{K_{\text{max}}}, \quad i \in \{\text{DL}, \text{UL}\}. \quad (2.3)$$

When the maximum number of users is active, the average load is simply given by the ratio of active to total number of time slots.

2.1 Communication Model

We now describe the communication at one data subcarrier ($q \in \{1, \dots, Q_{\text{DL}}\}$ in DL or $q \in \{1, \dots, Q_{\text{UL}}\}$ in UL) and one active OFDM symbol ($n \in \{1, \dots, N_{a,\text{DL}}\}$ in DL or $n \in \{1, \dots, N_{a,\text{UL}}\}$ in UL) to ease the notation. In the following, we specify when needed the dependency on q and n . All the quantities are expressed in the frequency domain as we consider them before the inverse fast Fourier transform (IFFT) at the transmitter side and after the fast Fourier transform (FFT) at the receiver side.

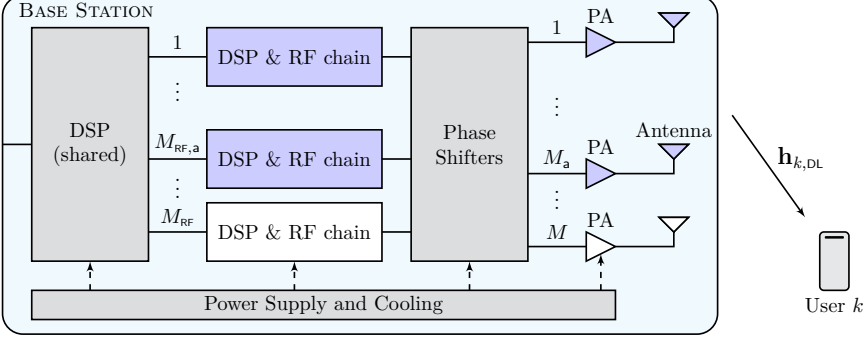


Figure 2.3: DL system model of this thesis, where the UL is symmetric except for the PAs being replaced by the LNAs and the channel being $\mathbf{h}_{k,\text{UL}}$. Purple color indicates an activated resource, while white indicates a deactivated resource.

In the thesis, we consider that M_a over M PAs and antennas are active in DL, and M_a over M LNAs and antennas are active in UL.⁵ Similarly, $M_{\text{RF},a}$ over M_{RF} are active in both DL and UL. Fig. 2.3 illustrates the DL system model. As assumed for the users and the time slots, the set of active antennas (RF chains) is represented by the first M_a ($M_{\text{RF},a}$) indexes over M (M_{RF}).

2.1.1 Downlink

The DL channel between the active BS antennas and the user k is denoted by $\mathbf{h}_{k,\text{DL}} \in \mathbb{C}^{M_a \times 1}$, hence the channel matrix is given by

$$\mathbf{H}_{\text{DL}} = [\mathbf{h}_{1,\text{DL}}, \dots, \mathbf{h}_{K_{\text{DL}},\text{DL}}]^T \in \mathbb{C}^{K_{\text{DL}} \times M_a}. \quad (2.4)$$

We assume a block-fading channel where the channel is constant over a coherence block that spans the coherence bandwidth and the coherence time [48]. The symbols of the users in DL are represented by the vector $\mathbf{s}_{\text{DL}} = [s_{1,\text{DL}}, \dots, s_{K_{\text{DL}},\text{DL}}]^T \in \mathbb{C}^{K_{\text{DL}} \times 1}$. The symbols are zero mean, uncorrelated among users, and with covariance equal to $\mathbf{D}_{\text{pDL}} = \text{diag}(p_{1,\text{DL}}, \dots, p_{K_{\text{DL}},\text{DL}})$, where $p_{k,\text{DL}}$ is the DL power allocation of user k . This can be compactly expressed as

$$\mathbb{E} \{\mathbf{s}_{\text{DL}} \mathbf{s}_{\text{DL}}^H\} = \mathbf{D}_{\text{pDL}}. \quad (2.5)$$

⁵We consider the antennas to be passive elements, hence we do not model their power consumption. With the term ‘active’ or ‘activated’ antenna, we indicate an antenna that is utilized for transmission or reception during the frame, in contrast to a ‘non-active’ or ‘deactivated’ antenna.

The BS performs hybrid analog-digital precoding and transmits the signal

$$\mathbf{x} = \mathbf{W}_{\text{ana}} \mathbf{W}_{\text{dig}} \mathbf{s}_{\text{DL}} \in \mathbb{C}^{M_a \times 1} \quad (2.6)$$

where $\mathbf{W}_{\text{ana}} \in \mathbb{C}^{M_a \times M_{\text{RF},a}}$ and $\mathbf{W}_{\text{dig}} = [\mathbf{w}_{\text{dig},1}, \dots, \mathbf{w}_{\text{dig},K_{\text{DL}}}] \in \mathbb{C}^{M_{\text{RF},a} \times K_{\text{DL}}}$ are the analog beamforming and digital precoding matrices, respectively. The vector $\mathbf{w}_{\text{dig},k} \in \mathbb{C}^{M_{\text{RF},a} \times 1}$ is the digital precoding vector of user k and has both a magnitude and a phase component. The analog beamforming matrix $\mathbf{W}_{\text{ana}}$, instead, contains phase-only elements and is selected from a codebook $\mathcal{W}_{\text{ana}}$.⁶ Hybrid analog-digital beamforming is a technique employed to reduce the number of RF chains compared to fully-digital beamforming [49]. More antennas share the same RF signal in hybrid beamforming, with reduction of BS power consumption. When there is no analog beamforming, the precoding is fully digital with $M_{\text{RF}} = M$, $M_{\text{RF},a} = M_a$ and $\mathbf{W}_{\text{ana}} = \mathbf{I}_{M_a}$, and the transmitted signal becomes

$$\mathbf{x} = \mathbf{W}_{\text{dig}} \mathbf{s}_{\text{DL}} \in \mathbb{C}^{M_a \times 1}. \quad (2.7)$$

We indicate the total transmit power at the BS, across all subcarriers, as $P_{\text{T,DL}}$. Considering that the cyclic prefix is larger than the maximal channel delay spread, the signal received by user k is expressed as

$$\begin{aligned} r_{k,\text{DL}} &= \mathbf{h}_{k,\text{DL}}^\top \mathbf{x} + \nu_{k,\text{DL}} = \mathbf{h}_{k,\text{DL}}^\top \mathbf{W}_{\text{ana}} \mathbf{W}_{\text{dig}} \mathbf{s}_{\text{DL}} + \nu_{k,\text{DL}} \\ &= \underbrace{\mathbf{h}_{k,\text{DL}}^\top \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig},k} s_{k,\text{DL}}}_{\text{desired signal}} + \underbrace{\sum_{k'=1, k' \neq k}^{K_{\text{DL}}} \mathbf{h}_{k,\text{DL}}^\top \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig},k'} s_{k',\text{DL}}}_{\text{inter-user interference}} + \underbrace{\nu_{k,\text{DL}}}_{\text{noise}} \end{aligned} \quad (2.8)$$

where $\nu_{k,\text{DL}}$ is zero-mean circularly-symmetric complex-Gaussian (ZMCSCG) noise with variance $\sigma_{k,\text{DL}}^2$. The noise is considered uncorrelated at the different users, and the received signal vector collecting the individual received signals is

$$\mathbf{r}_{\text{DL}} = \mathbf{H}_{\text{DL}} \mathbf{x} + \boldsymbol{\nu}_{\text{DL}} = \mathbf{H}_{\text{DL}} \mathbf{W}_{\text{ana}} \mathbf{W}_{\text{dig}} \mathbf{s}_{\text{DL}} + \boldsymbol{\nu}_{\text{DL}} \in \mathbb{C}^{K_{\text{DL}} \times 1} \quad (2.9)$$

where $\boldsymbol{\nu}_{\text{DL}} \sim \mathcal{CN}(\mathbf{0}_{K_{\text{DL}}}, \mathbf{D}_{\boldsymbol{\sigma}_{\text{DL}}}^2)$ and $\mathbf{D}_{\boldsymbol{\sigma}_{\text{DL}}}^2 = \text{diag}(\sigma_{1,\text{DL}}^2, \dots, \sigma_{K_{\text{DL}},\text{DL}}^2)$. To express the signal-to-interference-plus-noise ratio (SINR) of user k , we need to compute the power of the desired signal and divide it by the sum of the power of the inter-user interference (IUI) and of the noise. By using identity (2.5), the power of the desired signal is given by

$$\mathbb{E} \left\{ \left| \mathbf{h}_{k,\text{DL}}^\top \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig},k} s_{k,\text{DL}} \right|^2 \right\} = p_{k,\text{DL}} \left| \mathbf{h}_{k,\text{DL}}^\top \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig},k} \right|^2 \quad (2.10)$$

⁶Note that, while digital precoding is a pre-IFFT operation and it can therefore be applied per subcarrier, analog beamforming (i.e., the phase shifting) is a post-IFFT operation and is the same for each subcarrier.

while the power of the inter-user interference is expressed as

$$\mathbb{E}\left\{\left|\sum_{k'=1, k' \neq k}^{K_{\text{DL}}} \mathbf{h}_{k, \text{DL}}^{\text{T}} \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig}, k'} s_{k', \text{DL}}\right|^2\right\} = \sum_{k'=1, k' \neq k}^{K_{\text{DL}}} p_{k', \text{DL}} \left|\mathbf{h}_{k, \text{DL}}^{\text{T}} \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig}, k'}\right|^2. \quad (2.11)$$

Therefore, the DL SINR of user k can be written as

$$\text{SINR}_{k, \text{DL}} = \frac{p_{k, \text{DL}} \left|\mathbf{h}_{k, \text{DL}}^{\text{T}} \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig}, k}\right|^2}{\sum_{k'=1, k' \neq k}^{K_{\text{DL}}} p_{k', \text{DL}} \left|\mathbf{h}_{k, \text{DL}}^{\text{T}} \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig}, k'}\right|^2 + \sigma_{k, \text{DL}}^2}. \quad (2.12)$$

The above SINR is an instantaneous SINR for user k at one active subcarrier and active OFDM symbol. An achievable instantaneous rate per Hz can be expressed as

$$R_{k, \text{DL}} = \log_2 (1 + \text{SINR}_{k, \text{DL}}) \quad [\text{bit/s/Hz}]. \quad (2.13)$$

The sum rate is given by $\sum_{k=1}^{K_{\text{DL}}} R_{k, \text{DL}}$. The average rate over the frame (in bit/s) can be computed by averaging (empirically or probabilistically) expression (2.13) over the subcarriers and OFDM symbols, and multiplying it by the bandwidth B and the share of time over the frame duration in which data transmission occurs. This last multiplicative term is the usual pre-log factor that appears in the delivered rate expressions. In the following section and chapters, we adapt the rate or SINR expression to the considered scenario.

2.1.2 Uplink

In UL, the channel between the active BS antennas and the user k is given by $\mathbf{h}_{k, \text{UL}} \in \mathbb{C}^{M_{\text{a}} \times 1}$, therefore the channel matrix is denoted by

$$\mathbf{H}_{\text{UL}} = [\mathbf{h}_{1, \text{UL}}, \dots, \mathbf{h}_{K_{\text{UL}}, \text{UL}}] \in \mathbb{C}^{M_{\text{a}} \times K_{\text{UL}}}. \quad (2.14)$$

The users transmit the signal $\mathbf{s}_{\text{UL}} = [s_{1, \text{UL}}, \dots, s_{K_{\text{UL}}, \text{UL}}]^{\text{T}} \in \mathbb{C}^{K_{\text{UL}} \times 1}$, with covariance equal to $\mathbb{E}\{\mathbf{s}_{\text{UL}} \mathbf{s}_{\text{UL}}^{\text{H}}\} = \mathbf{D}_{\mathbf{p}_{\text{UL}}} = \text{diag}(p_{1, \text{UL}}, \dots, p_{K_{\text{DL}}, \text{UL}})$. The variable $p_{k, \text{UL}}$ represents the power transmitted by user k . The BS performs hybrid analog-digital combining via the matrices $\mathbf{V}_{\text{dig}} = [\mathbf{v}_{\text{dig}, 1}, \dots, \mathbf{v}_{\text{dig}, K_{\text{UL}}}]^{\text{T}} \in \mathbb{C}^{K_{\text{UL}} \times M_{\text{RF}, \text{a}}}$ and $\mathbf{V}_{\text{ana}} \in \mathbb{C}^{M_{\text{RF}, \text{a}} \times M_{\text{a}}}$, where $\mathbf{V}_{\text{ana}}$ is selected from a phase-only codebook $\mathcal{V}_{\text{ana}}$. The digital combining vector of user k is denoted by $\mathbf{v}_{\text{dig}, k} \in \mathbb{C}^{M_{\text{RF}, \text{a}} \times 1}$. After the users' transmission took place, the BS first receives the signal

$$\mathbf{r}_{\text{UL}} = \mathbf{H}_{\text{UL}} \mathbf{s}_{\text{UL}} + \tilde{\boldsymbol{\nu}}_{\text{UL}} \in \mathbb{C}^{M_{\text{a}} \times 1} \quad (2.15)$$

and then obtains the symbol estimate of user k as

$$y_{k, \text{UL}} = \mathbf{v}_{\text{dig}, k}^{\text{T}} \mathbf{V}_{\text{ana}} \mathbf{H}_{\text{UL}} \mathbf{s}_{\text{UL}} + \nu_{k, \text{UL}} \quad (2.16)$$

where $\nu_{k,\text{DL}} = \mathbf{v}_{\text{dig},k}^T \mathbf{V}_{\text{ana}} \tilde{\mathbf{v}}_{\text{UL}}$, with variance $\sigma_{k,\text{UL}}^2$, and $\tilde{\mathbf{v}}_{\text{UL}} \sim \mathcal{CN}(\mathbf{0}_{K_{\text{UL}}}, \mathbf{D}_{\tilde{\sigma}_{\text{UL}}}^2)$ is ZMCSCG noise with covariance matrix $\mathbf{D}_{\tilde{\sigma}_{\text{UL}}}^2 = \text{diag}(\tilde{\sigma}_{1,\text{UL}}^2, \dots, \tilde{\sigma}_{K_{\text{UL}},\text{UL}}^2)$. Following the same steps as for the DL, the instantaneous UL SINR for user k can be written as

$$\text{SINR}_{k,\text{UL}} = \frac{p_{k,\text{UL}} |\mathbf{v}_{\text{dig},k}^T \mathbf{V}_{\text{ana}} \mathbf{h}_{k,\text{UL}}|^2}{\sum_{k'=1, k' \neq k}^{K_{\text{UL}}} p_{k',\text{UL}} |\mathbf{v}_{\text{dig},k}^T \mathbf{V}_{\text{ana}} \mathbf{h}_{k',\text{UL}}|^2 + \sigma_{k,\text{UL}}^2} \quad (2.17)$$

and the corresponding achievable rate per Hz is

$$R_{k,\text{UL}} = \log_2(1 + \text{SINR}_{k,\text{UL}}) \quad [\text{bit/s/Hz}]. \quad (2.18)$$

As for the DL communication, we adapt in the following the rate or SINR expression to the analyzed scenario.

2.2 Power Consumption Model

In this section, we describe the power model used in the thesis. To evaluate the energy performance of BSs and design algorithms to optimize it, accurate power models are essential. A review of the models published in the literature until 2023 is given in [50]. The authors propose to classify the models in (i) linear models if they express the power consumption as linearly dependent on the BS output power, (ii) subcomponent models if they estimate the consumption of individual processing blocks, and (iii) massive multiple-input multiple-output (MIMO) models when they are specific to BSs with many antennas (e.g., 32 or more). The models typically target a specific frequency band (FR1 or FR2) and are used by both academic researchers and companies to perform evaluations, also at a standardized level [30]. Among more recent works, model [15] is a parametric power model that captures the power consumption of FR1 fourth-generation (4G) and fifth-generation (5G) BSs in Belgium. By utilizing measurements from a Belgian mobile network operator (MNO) and datasheets of BS manufacturers, the authors model the consumption of PAs, analog front-end (AFE), digital baseband (DBB), and power supply and cooling systems. The model applies to both massive MIMO and small-dimension MIMO BSs, and its parameters allow the inclusion of different energy-saving features such as micro-discontinuous transmission (μDTX). The PA model includes a non-constant PA efficiency and is discussed in Section 2.2.1. The model does not include subcomponent modeling of AFE and DBB, which are also considered independent from the DL physical resource load $\overline{\ell}_{\text{DL}}$ (differently from the PA) and the UL physical resource load $\overline{\ell}_{\text{UL}}$. The average BS consumption over the frame is computed by averaging the consumption of PA, AFE, and DBB over different operating modes, from working to sleep mode.

The model in [51], instead, is specific to FR2 and THz BSs and builds upon previous FR1 models in [19] and related papers. The main difference with [15] lies in the subcomponent modeling of AFE and DBB. Scaling laws are provided for analog HW devices such as digital-to-analog converters (DACs) and analog-to-digital converters (ADCs), as well as for digital HW modules such as MIMO precoders and combiners. In the model proposed hereinafter, we combine both approaches and build a power model that has component-level frame averaging over different operating modes with subcomponent modeling during working mode. We then showcase its application to BSs operating at FR3, which is receiving growing interest toward 6G networks [43].

Remark: We adopt the PA power model and the frame-averaging over different operating modes—including the use of power-reduction factors and implying the load dependency—from the model in [15], while we adopt the power modeling approaches for digital and analog subcomponents from the model in [51]. The fundamental differences with respect to [15] are the power modeling of digital and analog subcomponents, their load dependency, and the link with communication performance (i.e., ergodic rates) considering hybrid beamforming. The main novelty is in the analysis of prospective FR3 BSs, which we expect to combine the different features of power and communication models presented here.

We subdivide the BS in *components* and *subcomponents*, where we consider three different components: PAs, analog signal processing, and digital signal processing. Each component has an associated supply and cooling module. The total BS power consumption averaged over the frame is expressed as

$$P_{\text{cons}} = \frac{\overline{P_{\text{PA}}}}{\eta_{\text{PA},s/c}} + \frac{\overline{P_{\text{ana}}}}{\eta_{\text{ana},s/c}} + \frac{\overline{P_{\text{dig}}}}{\eta_{\text{dig},s/c}} \quad (2.19)$$

where $\overline{P_{\text{PA}}}$, $\overline{P_{\text{ana}}}$, and $\overline{P_{\text{dig}}}$ are the average consumptions by the PAs, the analog processing, and the digital processing. The quantities $\eta_{\text{PA},s/c} \in (0, 1]$, $\eta_{\text{ana},s/c} \in (0, 1]$, and $\eta_{\text{dig},s/c} \in (0, 1]$ are the supply and cooling efficiencies of each component. In case of no active cooling, the supply and cooling efficiency equals 1. Fig. 2.4 summarizes the BS architecture that we consider in the thesis.

Every component comprises one or more subcomponents that can be *active* or *non-active* for the full frame duration, where an active subcomponent can be in (i) working mode, (ii) micro-sleep mode, or (iii) idle mode, while a non-active subcomponent is in sleep mode. We employ reduction factors $\delta_{\text{mode}} \in [0, 1]$ to quantify the power consumption reduction of a subcomponent in a non-working mode compared to its working mode consumption [15].⁷ We emphasize that

⁷These power-reduction factors are considered to be a fixed feature of the HW, determined at production time, hence do not depend on the load and their values are not optimized during operation. An ideal HW would have all reduction factors equal to zero (e.g., $\delta_{\text{PA},\text{micro}} = 0$ meaning that the PA consumes no power during micro-sleep mode).

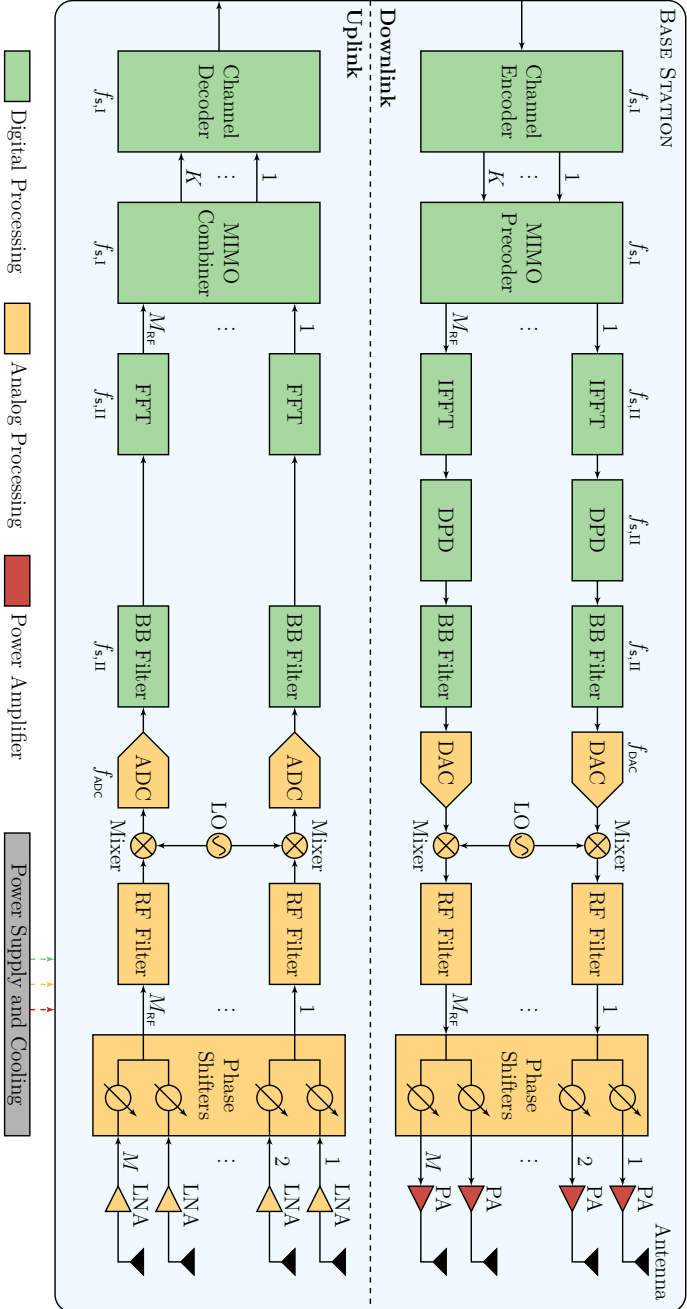


Figure 2.4: BS architecture considered in this thesis, where components are color coded, and a component comprises multiple subcomponents. Note that with fully-digital beamforming, the phase shifters are not present and $M_{RF} = M$. In this figure, active and non-active antennas or RF/DSP chains are not highlighted, and the BS serves K users in both DL and UL. The variables above or below some subcomponents are the sampling frequencies at which they operate.

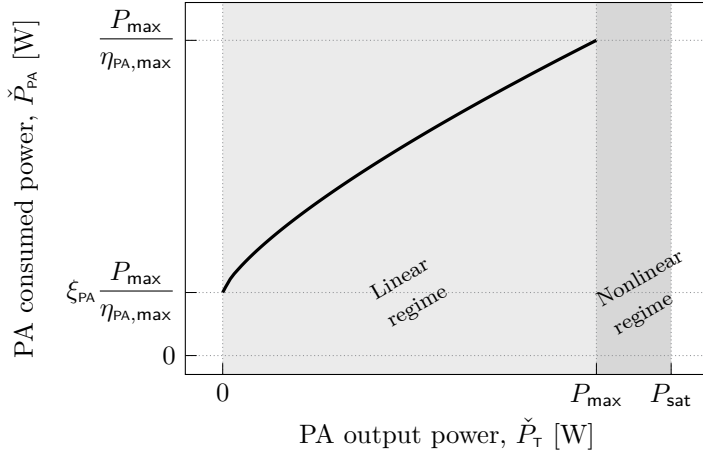


Figure 2.5: Illustration of PA power consumption versus PA output power, where in our model we consider $\check{P}_T \leq P_{\max}$ so that the PA is in the linear input-power output-power regime.

given the frame averaging of power consumption, the energy consumed by the BS in the frame is linearly proportional to the power consumed by the BS and given by $T_{\text{frame}}P_{\text{cons}}$. In the following, we describe the power consumption models of each component.

2.2.1 Power Amplifier

The PA has been and still is one of the devices consuming most of the power in cellular macro BSs due to the large BS-user distances and the consequent high transmit powers required. Evaluations on Long-Term Evolution (LTE) macro BSs have shown that the PA consumes more than 50% of the total BS consumption [14]. Model [15] has evaluated the PA power share to be at least 42% in New Radio (NR) macro BSs at full load. In the considered BS, there are M_a active PAs (and antennas) and $M - M_a$ non-active PAs (and antennas). The active PAs can be in three operating modes for the entire frame duration: working mode, micro-sleep mode, and idle mode. The non-active PAs are instead in sleep mode for the full frame duration.

The power consumption of a PA can be expressed as its output power divided by its efficiency. It is known that the PA efficiency, and hence the PA consumption, depends on the PA output power [40]. The larger the PA output power, the larger the PA efficiency. For an instantaneous output power $\check{P}_T$, the instantaneous PA

consumption is expressed via [15, eq. (4)] as

$$\check{P}_{\text{PA}}(\check{P}_{\text{T}}) = \xi_{\text{PA}} \frac{P_{\text{max}}}{\eta_{\text{PA,max}}} + (1 - \xi_{\text{PA}}) \frac{P_{\text{max}}^{1-\alpha} \check{P}_{\text{T}}^{\alpha}}{\eta_{\text{PA,max}}} \quad (2.20)$$

where P_{max} is the maximal PA output power considering an output power back-off from the PA saturation power P_{sat} , $\xi_{\text{PA}} \in [0, 1]$ controls the weight of static consumption to maximum consumption, $\eta_{\text{PA,max}} \in (0, 1]$ is the maximum PA efficiency, and $\alpha \in [0, 1]$. Fig. 2.5 illustrates the above function, where the back-off is usually represented by the ratio $P_{\text{max}}/P_{\text{sat}}$ and is in the range from -6 dB to -12 dB in OFDM systems due to the high peak-to-average power ratio (PAPR) [14]. Note that the maximal PA efficiency $\eta_{\text{PA,max}}$ is achieved when $\check{P}_{\text{T}} = P_{\text{max}}$. The parameters ξ_{PA} , α , and $\eta_{\text{PA,max}}$ can be tailored to specific PA architectures. Moreover, the maximal efficiency $\eta_{\text{PA,max}}$ and the saturation power P_{sat} typically decrease with the frequency [15, 52]. A PA power model similar to (2.20) is proposed in [25]. The works in [53, 54, 55] considered the same model with parameters $\xi_{\text{PA}} = 0$ and $\alpha = \frac{1}{2}$ for class B PAs, where the authors of [53, 54] further set $\eta_{\text{PA,max}} = \frac{\pi}{4}$, which is the theoretical maximal efficiency. The authors of [15] propose, for the sub-6 GHz band, to set $\xi_{\text{PA}} = 0$, $\alpha = \frac{1}{2}$, and $\eta_{\text{PA,max}} = 0.31$ to model class B PAs, and $\xi_{\text{PA}} \in (0, 1)$, $\alpha \in (0.5, 1)$, and $\eta_{\text{PA,max}} \in (0.08, 0.79)$ to model Doherty PAs. Based on two recent industrial PAs for 5G massive MIMO [56, 57], $\alpha = \frac{1}{2}$ proves to be a good fit to model their consumption in the linear regime⁸ (with a back-off of 6 dB), and the maximal efficiencies are $\eta_{\text{PA,max}} \in [0.21, 0.24]$ for the PA in the 3.3–3.8 GHz range [56], and $\eta_{\text{PA,max}} \in [0.34, 0.41]$ for the PA in the 4.4–5.0 GHz range [57]. In this thesis, we adapt the PA parameters in each chapter based on the considered PA technology and frequency band.

Let us consider that the instantaneous transmit power of a PA m in working mode during data transmission is denoted by $P_{\text{T},m}$. The consumption of the PA m is then given by

$$P_{\text{PA},m} = \check{P}_{\text{PA}}(P_{\text{T},m}) = \xi_{\text{PA}} \frac{P_{\text{max}}}{\eta_{\text{PA,max}}} + (1 - \xi_{\text{PA}}) \frac{P_{\text{max}}^{1-\alpha} P_{\text{T},m}^{\alpha}}{\eta_{\text{PA,max}}} \quad (2.21)$$

and the consumption of the M_{a} active PAs when in working mode is

$$P_{\text{PA}} = \sum_{m=1}^{M_{\text{a}}} P_{\text{PA},m}. \quad (2.22)$$

We instead consider that the M_{a} active PAs transmit at power of $\zeta_{\text{DL,sig}} P_{\text{max}}$ when they are in working mode during reference signaling, where $\zeta_{\text{DL,sig}} \in [0, 1]$.

⁸The linear regime refers to the range of output powers for which the input signal can be amplified without introducing nonlinear distortions.

The corresponding power consumption is $\check{P}_{\text{PA}}(\zeta_{\text{DL},\text{sig}} P_{\text{max}})$. The transmit power of an active PA during micro-sleep and idle mode is zero, and its consumption is therefore given by $\check{P}_{\text{PA}}(0)$ if no additional power-saving features are enabled. By recalling the frame structure in Fig. 2.2, the average PA consumption over the frame is computed by adding the consumption in each phase weighted by the duration of that phase, giving

$$\begin{aligned} \overline{P}_{\text{PA}} = & \underbrace{\overline{\ell}_{\text{DL}} \frac{K_{\text{max}}}{K_{\text{DL}}} \tau_{\text{DL}} (1 - \tau_{\text{DL},\text{sig}}) P_{\text{PA}}}_{\text{working mode, data transm.}} + M_{\text{a}} \underbrace{\left(\tau_{\text{DL}} \tau_{\text{DL},\text{sig}} \check{P}_{\text{PA}}(\zeta_{\text{DL},\text{sig}} P_{\text{max}}) \right)}_{\text{working mode, ref. signal transm.}} \\ & + \underbrace{\left(1 - \overline{\ell}_{\text{DL}} \frac{K_{\text{max}}}{K_{\text{DL}}} \right) \tau_{\text{DL}} (1 - \tau_{\text{DL},\text{sig}}) \check{P}_{\text{PA}}(0) \delta_{\text{PA},\text{micro}}}_{\text{micro-sleep mode}} + \underbrace{(1 - \tau_{\text{DL}}) \check{P}_{\text{PA}}(0) \delta_{\text{PA},\text{idle}}}_{\text{idle mode}} \\ & + (M - M_{\text{a}}) \underbrace{\check{P}_{\text{PA}}(0) \delta_{\text{PA},\text{sleep}}}_{\text{sleep mode}} \end{aligned} \quad (2.23)$$

where P_{PA} is given in (2.22), and $\delta_{\text{PA},\text{micro}} \in [0, 1]$, $\delta_{\text{PA},\text{idle}} \in [0, 1]$, and $\delta_{\text{PA},\text{sleep}} \in [0, 1]$ quantify the power savings when a PA is in micro-sleep (DTX), idle mode, and sleep mode, respectively. We refer to them as *reduction factors*. A reduction factor of 1 (symmetrically 0) indicates no (symmetrically full) power saving with respect to working mode. Note that we expressed $(N_{\text{a,DL}}/N_{\text{DL}}) = \overline{\ell}_{\text{DL}}(K_{\text{max}}/K_{\text{DL}})$ to emphasize the dependency of the power consumption on the load $\overline{\ell}_{\text{DL}}$. In the case where we consider an equal transmit power across PAs in working mode and data transmission, denoted by P_{a} , expression (2.23) simplifies to

$$\begin{aligned} \overline{P}_{\text{PA}} = & M_{\text{a}} \left(\overline{\ell}_{\text{DL}} \tau_{\text{DL}} (1 - \tau_{\text{DL},\text{sig}}) \check{P}_{\text{PA}}(P_{\text{a}}) + \tau_{\text{DL}} \tau_{\text{DL},\text{sig}} \check{P}_{\text{PA}}(\zeta_{\text{DL},\text{sig}} P_{\text{max}}) \right) \\ & + (1 - \overline{\ell}_{\text{DL}}) \tau_{\text{DL}} (1 - \tau_{\text{DL},\text{sig}}) \check{P}_{\text{PA}}(0) \delta_{\text{PA},\text{micro}} + (1 - \tau_{\text{DL}}) \check{P}_{\text{PA}}(0) \delta_{\text{PA},\text{idle}} \\ & + (M - M_{\text{a}}) \check{P}_{\text{PA}}(0) \delta_{\text{PA},\text{sleep}}. \end{aligned} \quad (2.24)$$

It can be noticed as the average PA consumption has a linear dependence on the DL load $\overline{\ell}_{\text{DL}}$, and a nonlinear dependence on the transmit power P_{a} or $\check{P}_{\text{T},m}$ except in the case where the PA exponent α is equal to one. The case $\alpha = 1$ represents an ideal PA, whose consumption scales linearly with the transmit power. More details on this aspect are discussed in Chapter 3.

2.2.2 Analog Signal Processing

We consider that the analog signal processing includes the RF chains, comprising also DACs in DL and ADCs in UL, that process the signals after baseband processing until the PA input in DL and after antenna reception until baseband processing in UL. The phase shifters for analog beamforming are part of the analog processing. The subcomponents that are part of analog processing are illustrated in orange in Fig. 2.4. Note that the local oscillator (LO) is shared among the RF chains and the LNA is included in the UL subcomponents. In the evaluations, we will focus on hybrid partially-connected beamforming architectures with $M_{\text{PS}} = M/M_{\text{RF}}$ phase shifters per RF chain (equal to the number of antennas per subarray) and fully-digital beamforming architectures with $M_{\text{PS}} = 0$ phase shifters. We recall that we consider M_{a} RF chains in DL, of which $M_{\text{RF,a}}$ are active, and M_{a} RF chains in UL, of which $M_{\text{RF,a}}$ are active. The average consumption by analog processing over the frame is given by

$$\overline{P_{\text{ana}}} = \overline{P_{\text{ana,DL}}} + \overline{P_{\text{ana,UL}}} + P_{\text{ana,misc}} \quad (2.25)$$

where $\overline{P_{\text{ana,DL}}}$ and $\overline{P_{\text{ana,UL}}}$ are the average consumptions in DL and UL. To compute the consumption in DL or UL (represented by the variable $i \in \{\text{DL}, \text{UL}\}$), we follow the same approach as for the PA, giving

$$\begin{aligned} \overline{P_{\text{ana},i}} &= M_{\text{RF,a}} \left(\overline{\ell}_i \frac{K_{\text{max}}}{K_i} \tau_i (1 - \tau_{i,\text{sig}}) P_{\text{ana},i} + \tau_i \tau_{i,\text{sig}} P_{\text{ana},i} \right. \\ &\quad \left. + \left(1 - \overline{\ell}_i \frac{K_{\text{max}}}{K_i} \right) \tau_i (1 - \tau_{i,\text{sig}}) P_{\text{ana},i} \delta_{\text{ana,micro}} + (1 - \tau_i) P_{\text{ana},i} \delta_{\text{ana,idle}} \right) \\ &\quad + (M_{\text{RF}} - M_{\text{RF,a}}) P_{\text{ana},i} \delta_{\text{ana,sleep}}, \quad i \in \{\text{DL}, \text{UL}\} \end{aligned} \quad (2.26)$$

where $\delta_{\text{ana,micro}} \in [0, 1]$, $\delta_{\text{ana,idle}} \in [0, 1]$, and $\delta_{\text{ana,sleep}} \in [0, 1]$ are reduction factors when the analog subcomponents are in micro-sleep, idle, and sleep mode, respectively. The term $P_{\text{ana,misc}}$ is a fixed consumption term that accounts for miscellaneous contributions other than those of the considered subcomponents. The average analog consumption depends linearly on both the DL and UL loads $\overline{\ell}_{\text{DL}}$ and $\overline{\ell}_{\text{UL}}$, and also linearly on $M_{\text{RF,a}}$. The total analog processing consumption during working mode in DL and UL is computed as

$$P_{\text{ana,DL}} = \frac{P_{\text{LO}}}{M_{\text{RF,a}}} + 2P_{\text{DAC}} + 2P_{\text{filter,RF}} + 2P_{\text{mixer}} + M_{\text{PS}} P_{\text{PS}} \quad (2.27)$$

$$P_{\text{ana,UL}} = \frac{P_{\text{LO}}}{M_{\text{RF,a}}} + 2P_{\text{ADC}} + 2P_{\text{filter,RF}} + 2P_{\text{mixer}} + M_{\text{PS}} P_{\text{PS}} + \frac{M_{\text{a}}}{M_{\text{RF,a}}} P_{\text{LNA}} \quad (2.28)$$

where P_{LO} , P_{DAC} , P_{ADC} , $P_{\text{filter,RF}}$, P_{mixer} , P_{PS} , and P_{LNA} are the power consumptions of an LO, DAC, ADC, RF filter, mixer, phase shifter, and LNA, respectively.

We include two DACs, ADCs, RF filters, and mixers per RF chain, processing the in-phase (I) and quadrature (Q) channels. Note that the LO consumption is divided by $M_{\text{RF},a}$ because the LO is shared among RF chains, while in $P_{\text{ana,UL}}$ the LNA contribution scales as the number of antennas M_a because there are as many LNAs as antennas (cf. Fig. 2.4).

Let us now describe the power models of subcomponents. In the first part of the evaluations, we will consider that the DAC and the ADC operate at large sampling frequencies (in the order of Gsample/s) to enable full flexibility in the digital processing module. This follows the recent trend of RF-DACs and RF-ADCs [58, 59, 60]. A model for the DAC power consumption is given by [61]

$$P_{\text{DAC}} = \Xi_{\text{DAC},1} 2^{b_{\text{DAC}}} + \Xi_{\text{DAC},2} b_{\text{DAC}} f_{\text{DAC}} \quad (2.29)$$

where b_{DAC} is the effective number of bits, f_{DAC} is the sampling frequency, and $\Xi_{\text{DAC},1}$, $\Xi_{\text{DAC},2}$ are parameters. This well models state-of-the-art high-speed DACs such as [62]. The ADC power consumption can be generally modeled as [63]

$$P_{\text{ADC}} = \Xi_{\text{ADC}} f_{\text{ADC}} 2^{b_{\text{ADC}}} \quad (2.30)$$

where Ξ_{ADC} is the Walden's figure-of-merit, b_{ADC} is the effective number of bits, and f_{ADC} is the sampling frequency. We express the consumption of a mixer, phase shifter, and LNA as [64, 65, 66]

$$P_{\text{mixer}} = \Xi_{\text{mixer}} f_{c,i}, \quad P_{\text{PS}} = \Xi_{\text{PS}} B, \quad P_{\text{LNA}} = \Xi_{\text{LNA}} B \quad (2.31)$$

where Ξ_{mixer} , Ξ_{PS} , and Ξ_{LNA} are figure-of-merit parameters that depend on gain, noise factor, and resolution. The consumptions of the LO and of the RF filter, represented by P_{LO} and $P_{\text{filter,RF}}$, are considered to be constants. In Section 2.3, we use manufacturers' datasheets to set their values.

2.2.3 Digital Signal Processing

The digital signal processing part of a BS receives the information bits and returns the input to the DACs in DL, while it receives the ADCs output and returns the information bits in UL. The digital subcomponents that we consider in this thesis are illustrated in green in Fig. 2.4. The frame-averaged digital consumption in both DL and UL is quantified as

$$\overline{P_{\text{dig}}} = \overline{P_{\text{dig,DL}}} + \overline{P_{\text{dig,UL}}} + P_{\text{dig,link}} \quad (2.32)$$

where the digital consumption in DL or UL averaged over the frame is

$$\begin{aligned} \overline{P_{\text{dig},i}} &= M_{\text{RF},a} \left(\overline{\ell}_i \frac{K_{\text{max}}}{K_i} \tau_i (1 - \tau_{i,\text{sig}}) P_{\text{dig},i} + \tau_i \tau_{i,\text{sig}} P_{\text{dig},i} \right. \\ &\quad \left. + \left(1 - \overline{\ell}_i \frac{K_{\text{max}}}{K_i} \right) \tau_i (1 - \tau_{i,\text{sig}}) P_{\text{dig},i} \delta_{\text{dig},\text{micro}} + (1 - \tau_i) P_{\text{dig},i} \delta_{\text{dig},\text{idle}} \right) \\ &\quad + (M_{\text{RF}} - M_{\text{RF},a}) P_{\text{dig},i} \delta_{\text{dig},\text{sleep}}, \quad i \in \{\text{DL}, \text{UL}\} \end{aligned} \quad (2.33)$$

with $\delta_{\text{dig},\text{micro}} \in [0, 1]$, $\delta_{\text{dig},\text{idle}} \in [0, 1]$, and $\delta_{\text{dig},\text{sleep}} \in [0, 1]$ being the reduction factors when the digital subcomponents are in micro-sleep, idle, and sleep mode, respectively. The term $P_{\text{ref},\text{link}}$ is a fixed consumption term accounting for link-layer consumption. As for the analog processing, the digital processing consumption scales linearly with the loads and the number of active and total RF chains, and does not depend on the transmit power. The consumption of an active DL or UL DSP chain is given by

$$P_{\text{dig},\text{DL}} = P_{\text{encoder}} + P_{\text{precoder}} + P_{\text{IFFT}} + P_{\text{DPD}} + P_{\text{filter,BB}} \quad (2.34)$$

$$P_{\text{dig},\text{UL}} = P_{\text{decoder}} + P_{\text{combiner}} + P_{\text{FFT}} + P_{\text{filter,BB}} \quad (2.35)$$

where P_{encoder} , P_{decoder} , P_{precoder} , P_{combiner} , P_{IFFT} , P_{FFT} , $P_{\text{filter,BB}}$ are the power consumptions of channel encoder and decoder, MIMO precoder and combiner, IFFT and FFT for OFDM modulation and demodulation, digital pre-distortion (DPD), and baseband (BB) filter.

To quantify the power consumption of a subcomponent, we include a static (signal-independent) and a dynamic (signal-dependent) term, which are influenced by the HW architecture, circuit, and semiconductor technology implementation. The static term is indicated by the subscript ‘s’, while following the approach in [19, 51], the dynamic term is computed as the number of complex giga operations per second (GOPS) divided by the computational efficiency (indicated by $\eta_{\text{component}}$) in complex GOPS/W. The channel encoding/decoding and the MIMO precoding/combining operate at the constellation rate, denoted as $f_{s,\text{I}}$, while the IFFT, FFT, DPD, and BB filtering operate at a sampling rate larger than $f_{s,\text{I}}$, denoted as $f_{s,\text{II}}$.⁹

Let us express the power consumption of each subcomponent when in active mode, i.e., when transmitting or receiving data or reference signals. The consumption of the channel encoder and decoder is computed as

$$P_j = \frac{P_{j,s}}{M_{\text{RF},a}} + \frac{f_{s,\text{I}}}{\eta_j} \frac{\Xi_j}{M_{\text{RF},a}}, \quad j \in \{\text{encoder}, \text{decoder}\} \quad (2.36)$$

⁹The two sampling rates are equal to $f_{s,\text{I}} = \mu_s B$, where $\mu_s \in [0, 1]$ and B is the bandwidth, and $f_{s,\text{II}} = Q_{\text{IFFT}} \Delta_f$, where Q_{IFFT} is the IFFT size and Δ_f is the subcarrier spacing.

where $P_{j,s}$ is the static consumption among all the $M_{\text{RF},a}$ active DSP chains (i.e., all the digital signal paths that are not switched off), η_j is the computational efficiency, and Ξ_j is the number of complex GOPS per sample among all the $M_{\text{RF},a}$ active DSP chains. Similarly, the consumption of MIMO precoder and combiner is quantified by

$$P_j = \frac{P_{j,s}}{M_{\text{RF},a}} + \frac{f_{s,I}}{\eta_j} \Xi_j, \quad j \in \{\text{precoder, combiner}\} \quad (2.37)$$

where this time Ξ_{precoder} and Ξ_{combiner} give the number of complex GOPS per sample and per active DSP chain. The same formula as above applies for the IFFT, FFT, DPD, and BB filter, with the only difference being the different sampling frequency $f_{s,\text{II}}$. This gives

$$P_j = \frac{P_{j,s}}{M_{\text{RF},a}} + \frac{f_{s,\text{II}}}{\eta_j} \Xi_j, \quad j \in \{\text{IFFT, FFT, DPD, filter, BB}\}. \quad (2.38)$$

To give an example, let us consider that a zero-forcing (ZF) scheme is used for MIMO precoding and combining. Assuming channel reciprocity between DL and UL, i.e., $\mathbf{H}_{\text{DL}} = \mathbf{H}_{\text{UL}}^H$, one can set $\mathbf{W}_{\text{ana}} = \mathbf{V}_{\text{ana}}^H$. The digital precoding and decoding matrix are given by $\mathbf{W}_{\text{dig}} = \mathbf{V}_{\text{dig}}^H = \mathbf{H}_{\text{eff}}^H (\mathbf{H}_{\text{eff}} \mathbf{H}_{\text{eff}}^H)^{-1}$, where $\mathbf{H}_{\text{eff}} = \mathbf{H}_{\text{DL}} \mathbf{W}_{\text{ana}}$ is the effective channel matrix seen by the digital precoder/combiner. Considering that the matrix inversion is computed via Cholesky decomposition and backward-forward substitution, the number of complex GOPS per sample and per DSP chain required by the precoder and the combiner are [21]

$$\Xi_{\text{precoder}} = 2K + \frac{1}{\iota_{\text{coh}}} \left(\frac{K^3}{3M_{\text{RF},a}} + 3K^2 + K \right), \quad \Xi_{\text{combiner}} = 2K \quad (2.39)$$

where ι_{coh} is the number of samples per coherence block.

2.3 Application to Upper Mid-Band Base Stations

We present here an application of the power and communication models to prospective upper mid-band BSs. This band, also known as FR3, is expected to trade off the benefits of FR1 (or sub-7 GHz band) in terms of propagation and coverage, with the large bandwidths offered by FR2 (or mmWave band) [44]. Recent works are characterizing FR3 with focus on aspects such as channel propagation [67], measurements of indoor channels [68], prototyping and channel sounders [69], and new fully-digital MIMO arrays having (much) more antenna elements than in current massive MIMO [70]. Evaluating the power consumption of FR3 BSs constitutes a fundamental task to link the

communication performance with environmental and economic sustainability. In this regard, the authors of [71] have studied the capacity and power consumption of networks using both FR1 and FR3. The power consumption model that they consider does not include non-working modes of components, hence it does not quantify the load dependency, nor subcomponents modeling. Further, we point out that applying models [15] and [51] to FR3 BSs is not straightforward. Models [15], although based on measurements at FR1, relies on fixed reference values for aggregate analog and digital consumption. Scaling these figures to a BS array with, e.g., 1024 antennas would introduce significant uncertainty. Model [51], which targets FR2, assumes RF technologies and system parameters that differ from those envisioned for FR3 (e.g., Indium phosphide (InP) PAs and bandwidths above 1 GHz). In the following, we present the HW architectural choices and the evaluations in the considered FR3 scenarios.

2.3.1 Architectural Choices

Let us focus on an upper mid-band BS with 1024 antennas and PAs in DL, and 1024 antennas and LNAs in UL. The number of RF chains is varied from 16 to 1024, where the latter case represents fully-digital beamforming. All the former cases represent hybrid partially-connected beamforming. MIMO configurations with hundreds to thousands of antenna elements and RF chains are being proposed to provide cell capacity boost in the 6G upper mid-band [45]. We consider a case in which the PAs/LNAs and RF/DSP chains are active for the full frame duration, giving $M_a = M$ and $M_{\text{RF},a} = M_{\text{RF}}$. This reduces the number of parameters and allows us to more clearly compare the selected BS architectures. We also assume, as simplification, that $P_{\text{ana,misc}} = P_{\text{dig,link}} = 0$. The bandwidth is fixed to $B = 400$ MHz and the number of users is $K_{\text{max}} = K_{\text{DL}} = K_{\text{UL}} = K = 8$. We consider that the BS operates in time-division duplexing (TDD) with $\tau_{\text{DL}} = 0.75$ and $\tau_{\text{UL}} = 0.25$. The carrier frequencies are set as $f_{\text{c,DL}} = f_{\text{c,UL}} = 10$ GHz, the subcarrier spacing as $\Delta_f = 120$ kHz, the IFFT size as $Q_{\text{IFFT}} = 4096$, and the number of data subcarriers as $Q_{\text{max}} = 3000$.

Power Amplifier

Concerning the PA, the four main technologies for implementing PAs at FR3 are gallium nitride (GaN), gallium arsenide (GaAs), silicon-germanium (SiGe), and complementary metal-oxide-semiconductor (CMOS) [46]. Each one gives a different maximal output power P_{max} and efficiency $\eta_{\text{PA,max}}$, among other factors. GaN PAs are typically characterized by larger maximal powers at the expense of lower efficiencies. We select a recent SiGe implementation for FR3, proposed in [72], and set the model parameters to $P_{\text{max}} = 0.1$ W, $\eta_{\text{PA,max}} = 0.15$, $\alpha = 0.75$,

and $\xi_{\text{PA}} = 0.1$. We consider an average per-PA (and per-antenna) transmit power of $P_a = P_{\text{T,DL}}/M$ W, where $P_{\text{T,DL}}$ is the total transmit power. The average PA consumption is therefore computed via equation (2.24).

Analog Processing

In the analog processing, we consider that both the DAC and the ADC operate at sampling frequencies of $f_{\text{DAC}} = f_{\text{ADC}} = 5$ Gsample/s and with effective number of bits equal to $b_{\text{DAC}} = b_{\text{ADC}} = 8$ [62, 73]. The model [61] quantifies the DAC's parameters as $\Xi_{\text{DAC},1} = 1.5 \cdot 10^{-5}$ and $\Xi_{\text{DAC},2} = 1.5 \cdot 10^{-12}$. Based on a state-of-the-art high-speed ADC [73], we set the Walden's figure-of-merit to $\Xi_{\text{ADC}} = 70$ fJ/conversion step. Recent implementations of a mixer, phase shifter, and LNA for FR3 give the respective parameters $\Xi_{\text{mixer}} = 2.5 \cdot 10^{-13}$, $\Xi_{\text{PS}} = 3.5 \cdot 10^{-11}$, and $\Xi_{\text{LNA}} = 2.7 \cdot 10^{-11}$ [64, 65, 66]. The datasheets of an LO [74] and RF filter [75] are used to set $P_{\text{LO}} = 40$ mW and $P_{\text{filter,RF}} = 5$ mW.

Remark: We caution that the power consumption of commercial HW may be larger than the one estimated here (resulting from designs published in research works). In a commercial HW solution, there could be other HW modules that consume power, in addition to the module under study. This extra power consumption can be included in the model as part of the (constant) parameter $P_{\text{ana,misc}}$ in equation (2.25).

Digital Processing

The two most common HW solutions for the baseband processing of cellular wireless BSs are field-programmable gate array (FPGA) and application-specific integrated circuit (ASIC). The first provides more flexibility in terms of reconfiguration at the cost of larger power consumption and lower speed. Each digital subcomponent can be associated with a specific HW platform in the power model by tailoring its parameters $P_{\text{subcomponent,s}}$ (static consumption) and $\eta_{\text{subcomponent}}$ (computational efficiency), cf. Section 2.2.3. To estimate the parameters of FPGA solutions, we synthesized, via high-level synthesis in the AMD VivadoTM environment, a MIMO precoder based on LDL decomposition on an AMD ZynqTM UltraScale+TM RF system-on-chip (RFSoc) ZCU216 [58]. With 32 antennas, 8 users, and a clock frequency of 200 MHz, the computations of precoding matrices and precoded signals are performed at 0.8 MHz and 400 MHz, respectively, where we estimate $\eta_{\text{precoder}} = 0.2 \cdot 10^3$ complex GOPS/W and a static consumption of 1 W. For ASIC implementations, we consider a ten times larger efficiency and a ten times lower static power consumption [76].

Remark: The $10\times$ difference in GOPS/W between ASIC and FPGA, although in line with a class of recent HW accelerators for neural networks [77], is an assumption and must be taken with care. Synthesis and implementation of ASIC solutions for communication tasks of FR3 BS are necessary to advance and refine this comparison.

By using the values provided in [51] for low-density parity-check (LDPC) codes, we set the parameters of the channel encoder and decoder to $\Xi_{\text{encoder}} = 0.59(R_{\text{DL}}/0.9B)$ and $\Xi_{\text{decoder}} = 29.17(R_{\text{UL}}/0.9B)$, where R_{DL} and R_{UL} are the DL and UL sum rates. We consider that both subcomponents are implemented in ASIC with $P_{\text{encoder},s} = P_{\text{decoder},s} = 0.1$ W and $\eta_{\text{encoder}} = \eta_{\text{decoder}} = 0.2 \cdot 10^4$ GOPS/W. As regards MIMO precoding and combining, we consider that ZF is used, requiring the GOPS per sample and per RF chain specified in equation (2.39). Due to the flexibility required to change the beamforming scheme with the scenario (e.g., interference-limited or noise-limited), we select FPGA as HW architecture, giving $\eta_{\text{precoder}} = \eta_{\text{combiner}} = 0.2 \cdot 10^3$ GOPS/W. If we consider that a single FPGA is shared among 32 RF chains, the static consumption is $P_{\text{precoder},s} = P_{\text{combiner},s} = \lceil M_{\text{RF},a}/32 \rceil \cdot 1$ W. For OFDM (de)modulation, we assume a 5G-like operation where the number of subcarriers is fixed and different subcarrier spacings determine the bandwidth [32]. We consider that they are implemented in ASIC with $P_{\text{IFFT},s} = P_{\text{FFT},s} = 0.1$ W and $\eta_{\text{IFFT}} = \eta_{\text{FFT}} = 0.2 \cdot 10^4$ GOPS/W. The GOPS of both IFFT and FFT are computed as $\Xi_{\text{IFFT}} = \Xi_{\text{FFT}} = \frac{3}{2} \log_2 Q_{\text{IFFT}}$. The DPD follows in the DL processing chain. It is used to linearize the overall input-output PA characteristic, and is typically characterized by large GOPS both for filtering and computing its coefficients. We utilize reference [78] to set $\Xi_{\text{DPD}} = 50$ and consider it is implemented in ASIC, giving $P_{\text{DPD},s} = 0.1$ W and $\eta_{\text{DPD}} = 0.2 \cdot 10^4$ GOPS/W. Last, the BB filter performs interpolation in DL and decimation in UL. Given that FR3 subbands are expected to have different spectral requirements, we opt for the FPGA solution [79].¹⁰ This yields $P_{\text{filter,bb},s} = \lceil M_{\text{RF},a}/32 \rceil \cdot 1$ W if we consider a single FPGA implementing 32 BB filters, and $\eta_{\text{filter,bb}} = 0.2 \cdot 10^3$ GOPS/W. Its GOPS is computed as $\Xi_{\text{filter,bb}} = N_{\text{filter,bb}} O_{\text{filter,bb}}$, where $N_{\text{filter,bb}}$ and $O_{\text{filter,bb}}$ are the number of filter taps and oversampling coefficient, which we set to $N_{\text{filter,bb}} = 20$ and $O_{\text{filter,bb}} = 4$.

2.3.2 Evaluation

In this section, we evaluate the power model and architectural choices together with the communication performance. We consider a 3GPP 39.801 Urban

¹⁰Considering the precoders, combiners, and BB filters to be implemented in FPGA induces a significant penalty in the power consumption (factor of 10 in our evaluations). In Section 2.3.3, we quantify the impact of this choice.

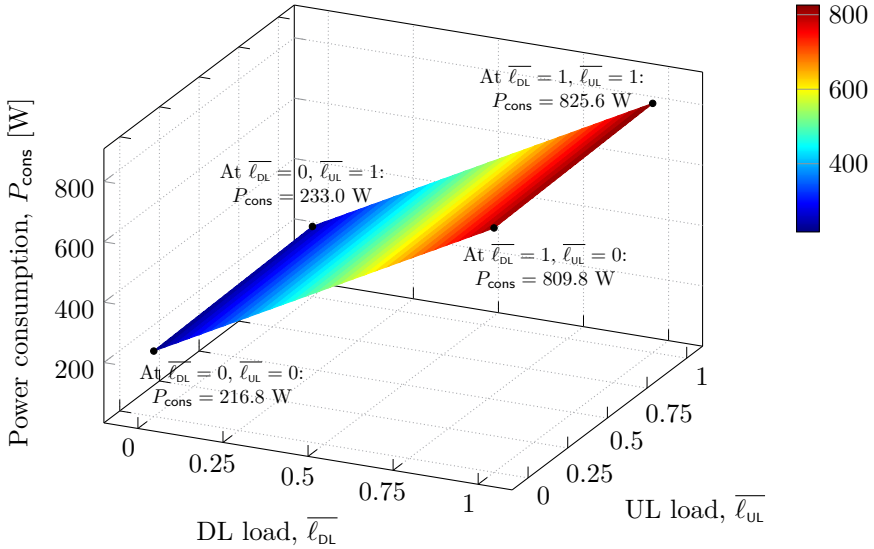


Figure 2.6: In the considered FR3 scenario and under the architectural choices of Section 2.3.1, BS power consumption versus DL and UL average physical resource loads. The main parameters are $M = 1024$ antennas, $M_{\text{RF}} = 256$ RF chains, $B = 400$ MHz, $K = 8$ users, $P_{\text{T,DL}} = 100$ W, $p_{k,\text{UL}} = 100$ mW $\forall k$, $\tau_{\text{DL}} = 0.75$, and $\tau_{\text{UL}} = 0.25$.

Micro (UMi) wideband channel model generated by the Sionna simulator [80]. The BS array is a uniform planar array (UPA) with 2048 antenna elements and half-wavelength antenna spacing in both directions. This gives $M = 1024$ in DL and $M = 1024$ in UL. The codebook for analog beamforming is the same in DL and UL, i.e., $\mathcal{W}_{\text{ana}} = \mathcal{V}_{\text{ana}}$, and is generated via a 2D-DFT. We set the total DL transmit power to $P_{\text{T,DL}} = 100$ W such that the PA output power is $P_{\text{a}} \approx 0.1 = P_{\text{max}}$ W. Further, we consider a uniform DL power allocation, giving $p_{k,\text{DL}} = 1$ $\forall k$. The UL transmit power is $p_{k,\text{UL}} = 100$ mW for all the users. The noise powers are set as $\sigma_{k,\text{DL}}^2 = \sigma_{k,\text{UL}}^2 = -123$ dBm $\forall k$. The parameters of the TDD protocol are $\tau_{\text{DL,sig}} = \tau_{\text{UL,sig}} = 1/14$ and $\zeta_{\text{DL,sig}} = 1/12$ [15]. We consider that channel reciprocity between DL and UL holds. We further set $\eta_{\text{dig,s/c}} = \eta_{\text{ana,s/c}} = \eta_{\text{PA,s/c}} = 0.8$. According to equation (2.3), the average physical loads are bounded as $\bar{\ell}_{\text{DL}} \in [0, 1]$ and $\bar{\ell}_{\text{UL}} \in [0, 1]$. Last, from equation (2.13) and equation (2.18), the ergodic sum

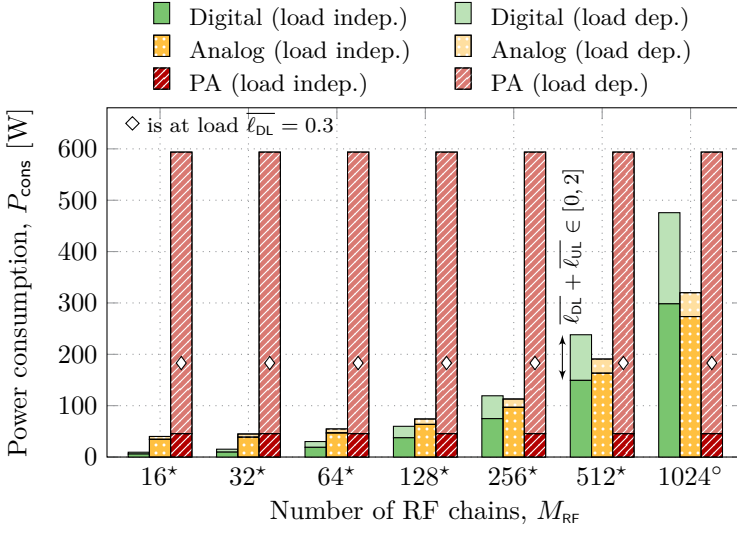
rates are computed as

$$\overline{R}_i = \overline{\ell}_i \tau_i (1 - \tau_{i,\text{sig}}) \tilde{B} \mathbb{E} \left\{ \sum_{k=1}^K R_{k,i} \right\} \quad [\text{bit/s}], \quad i \in \{\text{DL}, \text{UL}\} \quad (2.40)$$

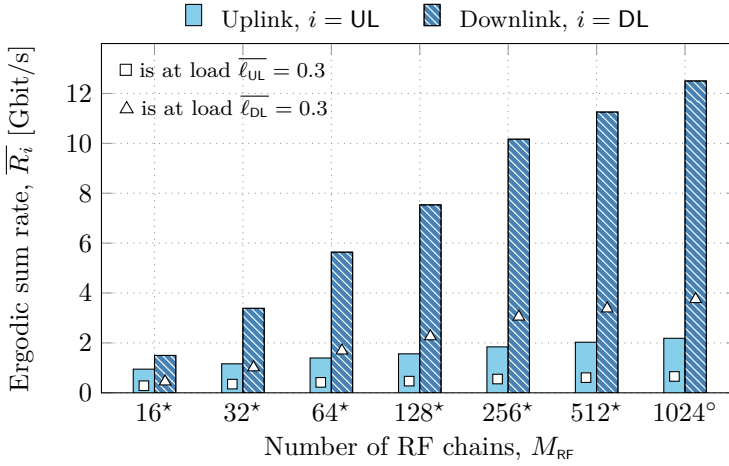
via Monte Carlo simulation, where $\tilde{B} = \mu_s B = 0.9 \cdot 400$ MHz is the effective bandwidth, accounting for the fact that 90% of the subcarriers contained in B are effectively used to transmit/receive the users' data. We consider that the PAs and the digital processing implement more time-domain power-saving features compared to the analog processing. Hence, we set the reduction factors of the PAs to $\delta_{\text{PA},\text{micro}} = 0.5$ and $\delta_{\text{PA},\text{idle}} = 0.25$, equally for the digital processing giving $\delta_{\text{dig},\text{micro}} = 0.5$ and $\delta_{\text{dig},\text{idle}} = 0.25$, while for the analog processing they are set to $\delta_{\text{ana},\text{micro}} = 0.75$ and $\delta_{\text{ana},\text{idle}} = 0.5$.

Fig. 2.6 illustrates the power consumption as a function of the DL and UL loads. The power consumption is an affine function of the loads. The DL load has a much higher impact on the consumption than the UL load, mainly due to the PAs consumption (that depends on $\overline{\ell}_{\text{DL}}$). At zero loads, the consumption is $P_{\text{cons}} = 216.8$ W, while it equals $P_{\text{cons}} = 825.6$ W at full loads. At typical loads (up to 30%), however, the consumption goes up to approximately 400 W, indicating that approximately 50% of the consumption is due to the static (load-independent) part. Note that, even though we do not analyze them here, energy-saving techniques would typically scale down the number of active antennas M_a , active RF chains $M_{\text{RF},a}$, and transmit powers $P_{\text{T,DL}}$ and $p_{k,\text{UL}}$ as a function of the loads, resulting in a non-affine (nonlinear) scaling of the power consumption at low loads.

Let us now quantify the impact of each BS component on the total consumption while varying the number of RF chains. In Fig. 2.7a, we show the power consumption and ergodic rates while varying the number of RF chains from 16 to 1024. The PA consumption is constant because it does not depend on M_{RF} but only on the number of antennas (and PAs) M , which is fixed to $M = 1024$. We split the consumption of each BS component in load dependent (lighter bars) and load independent (darker bars). The load-dependent terms are the ones in equation (2.23), (2.26), or (2.33) that multiply $\overline{\ell}_{\text{DL}}$ or $\overline{\ell}_{\text{UL}}$, and vice versa for the load-independent terms. We plot the full range of $\overline{\ell}_{\text{DL}} + \overline{\ell}_{\text{UL}}$, ranging from 0 to 2 and indicated with an arrow in one case. More realistic loads, up to $\overline{\ell}_{\text{UL}} = \overline{\ell}_{\text{DL}} = 0.3$, are highlighted with a marker. At full loads, the total BS consumption ranges from 643 W (when $M_{\text{RF}} = 16$) to 1389 W (when $M_{\text{RF}} = 1024$), while at $\overline{\ell}_{\text{UL}} = \overline{\ell}_{\text{DL}} = 0.3$, it varies from 253 W (when $M_{\text{RF}} = 16$) to 849 W (when $M_{\text{RF}} = 1024$). We observe that at full loads, the PA consumption is the largest among the BS components. At more realistic loads, however, the PA consumption becomes comparable to digital and analog consumptions approximately from $M_{\text{RF}} = 128$. The analog and digital processing have larger



(a)



(b)

*Hybrid analog-digital beamforming °Fully-digital beamforming

Figure 2.7: In the considered FR3 scenario and under the architectural choices of Section 2.3.1, (a) BS power consumption versus number of RF chains, (b) ergodic rates versus number of RF chains. For both figures, the main parameters are $M = 1024$ antennas, $B = 400$ MHz, $K = 8$ users, $P_{\text{T,DL}} = 100$ W, $p_{k,\text{UL}} = 100$ mW $\forall k$, $\tau_{\text{DL}} = 0.75$, and $\tau_{\text{UL}} = 0.25$.

Table 2.1: Energy efficiency versus number of RF chains at two load levels, referring to the architectural choices of Section 2.3.1.

M_{RF}	16	32	64	128	256	512	1024
EE at full loads [Mbit/J]	3.79	6.95	10.36	12.50	14.54	12.99	10.57
EE at 30% loads [Mbit/J]	2.89	5.21	7.50	8.51	9.01	7.15	5.19

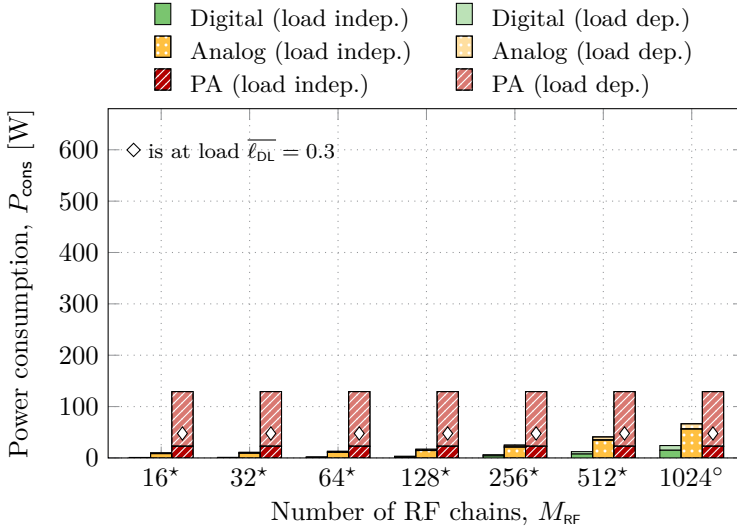
load-independent consumption compared to the load-dependent one, while it is the opposite for the PA. The largest contributors to the analog consumption are the LNAs at low values of M_{RF} , and the DACs and ADCs at high values of M_{RF} . In the digital consumption, the MIMO precoder/combiner and the BB filter, considered to be implemented in FPGA, have the largest share.

In Fig. 2.7b, we plot the ergodic rates in DL and UL for the full range of $\overline{\ell}_{\text{DL}}$ and $\overline{\ell}_{\text{UL}}$ (corresponding to the height of each bar), and highlight the values at loads of 30%. We point out that for small numbers of RF chains, the rates do not drastically drop because there are $M/M_{\text{RF}} = 1024/M_{\text{RF}}$ subarrays that perform analog beamforming. The DL sum rate reaches more than 12 Gbit/s and more than 2 Gbit/s in UL for the fully-digital beamforming at full load. We observe that hybrid beamforming with 128 or more RF chains provides rates larger than 50% of the ones achieved in both DL and UL by the fully-digital counterpart. The overall energy efficiency (EE), computed as $\text{EE} = (\overline{R}_{\text{DL}} + \overline{R}_{\text{UL}})/P_{\text{cons}}$, measured in bit per Joule and dependent on the loads, is shown in Table 2.1. The maximal EE of 12.5 Mbit/J is achieved by the hybrid beamforming configuration with 256 RF chains at full loads. The same configuration also improves the EE by a factor of $1.74\times$ at loads of 30%.

2.3.3 Impact of Architectural Choices

The HW architectural choices presented in Section 2.3.1 can have a significant impact on the final results. In this section, we focus on some critical assumptions, with consequent choice of parameters, and evaluate their effect on the evaluations. In particular, we consider a BS operating at a lower bandwidth and with a more energy-efficient choices for the DACs, ADCs, digital modules, and PAs. The introduced differences compared to Section 2.3.1 are the following:

- The bandwidth is scaled down to $B = 100$ MHz, instead of $B = 400$ MHz. The IFFT size is kept fixed at $Q_{\text{IFFT}} = 4096$ and the subcarrier spacing is reduced to $\Delta_f = 30$ kHz (instead of $\Delta_f = 120$ kHz). The lower bandwidth impacts both the analog and digital consumption. Accordingly,



*Hybrid analog-digital beamforming °Fully-digital beamforming

Figure 2.8: In the considered FR3 scenario and varying the architectural choices as explained in Section 2.3.3, BS consumption versus number of RF chains.

the transmit power is decreased to $P_{T,DL} = 25$ W and $p_{k,UL} = 25$ mW $\forall k$ (from $P_{T,DL} = 100$ W and $p_{k,UL} = 100$ mW $\forall k$).

- The DACs and ADCs operate at much lower sampling rates, i.e., $f_{DAC} = f_{ADC} = 500$ Msample/s, instead of $f_{DAC} = f_{ADC} = 5$ Gsample/s. This follows a more conventional operation of energy-optimized DACs and ADCs.
- The digital precoders, digital combiners, and BB filters are implemented in ASIC instead of FPGA. This gives a ten-times larger efficiency and ten-times lower static consumption, i.e., $\eta_{precoder} = \eta_{combiner} = \eta_{BB,filter} = 0.2 \cdot 10^4$ complex GOPS/W, and $P_{precoder,s} = P_{combiner,s} = P_{BB,filter,s} = 0.1$ W. As for the DACs and ADCs, this reduces the flexibility during operation but decreases the energy consumption.
- The PA efficiency doubles, giving $\eta_{PA,max} = 0.3$, instead of $\eta_{PA,max} = 0.15$. This foresees more energy-efficient PA implementations for FR3.

In Fig. 2.8, we plot the consumption against M_{RF} . Compared to Fig. 2.7a, there is a significant decrease in the power consumption by each BS component. Under fully-digital beamforming at full loads, the consumption of the PAs, analog processing, and digital processing decrease of 78%, 79%, and 95%, respectively.

Table 2.2: Energy efficiency versus number of RF chains at two load levels, referring to the architectural choices of Section 2.3.3.

M_{RF}	16	32	64	128	256	512	1024
EE at full loads [Mbit/J]	4.35	8.06	12.22	15.22	18.71	18.20	16.70
EE at 30% loads [Mbit/J]	2.85	5.23	7.80	9.44	11.03	9.93	8.36

These percentages of reduction are the same in the hybrid beamforming case with $M_{\text{RF}} = 256$. The bandwidth decrease has a larger impact on the digital consumption than on the analog consumption, which is instead more impacted by the reduced sampling frequencies of DACs and ADCs. In terms of EE (reported in Table 2.2) the hybrid beamforming configuration with $M_{\text{RF}} = 256$ RF chains remains the best option at both full loads and loads of 30%.

2.4 Conclusion

In this chapter, we have discussed the communication and power models used in the thesis. The BS architecture is split into PA, analog, and digital components, where we consider a number of active PAs as well as RF/DSP chains. Further, we consider that a number of active users are served by all the available data subcarriers during a number of active time slots, leaving the remaining time slots empty when not allocated for reference signaling. To provide a reply to **RQ1** on the power modeling of next-generation BSs, we proposed a generalization of existing power models by combining the frame-averaging of the components over different operating modes (from working to sleep) with subcomponent modeling that allows us to quantify the impact of individual HW devices or modules. Also, the load dependency of PAs is extended to analog and digital components to reflect upcoming BS implementations with load-dependent power-saving features in the RF/DSP chains other than in the PAs. We applied the power model to prospective upper mid-band or FR3 BSs and underlined the importance of considering communication performance to make the trade-off between power consumption and capacity transparent. Given the many open design questions on FR3 BSs, e.g., whether they should implement hybrid analog-digital or fully-digital beamforming, our analysis focused on quantifying both their impact on delivered rates and power consumption. In the set of evaluations, relative to an array of 1024 antenna elements using 400 MHz of bandwidth, we found that the hybrid analog-digital beamforming configuration with 256 RF chains provides the largest EE, reaching 18.71 Mbit/J.